Formula :

(1) Dynamic power = Capacitive load × Voltage^2 × Frequency

$= CV^2F$

(2) Static power = VI

(3) $T_{improved} = \frac{T_{affected}}{\text{improvement factor}} + T_{unaffected}$

↑ Amdahl

(4) $MIPS = \frac{\text{Instruction Count (IC)}}{\text{Execution time} \times 10^6}$

(5) $MIPS = \frac{\text{clock rate}}{CPI \times 10^6}$

↑ CPU Time $\propto \frac{1}{\text{performance} \downarrow}$

↑ CPU Time $= \frac{IC \times CPI}{f \downarrow}$

$f \propto$ performance

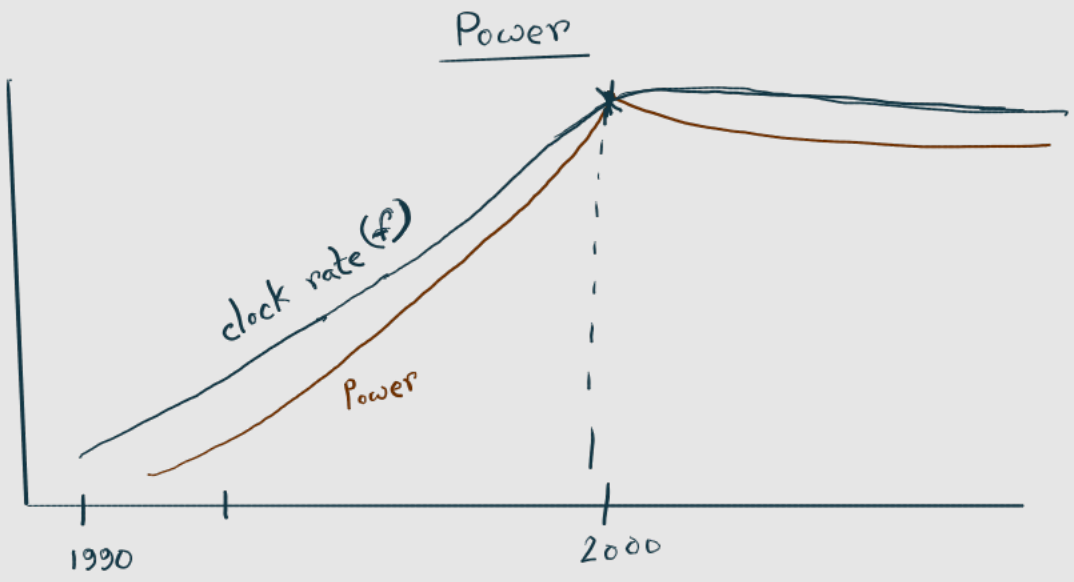
Better :

Performance ↑

CPU time ↓

Power ↓

Power = $L\ V^2\downarrow\ F\uparrow$



$$\text{CPU Time} = \frac{\text{Clock cycle}}{\text{Clock rate [f]}} \propto \frac{1}{\text{performance}}$$

$$\text{Dynamic power} = \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

$$= CV^2F$$

$$\text{Static power} = VI$$

Power wall : Can't reduce voltage any more. Heat can not be reduced as well, as it will cause unstability.

Solution : 1. Cache optimization
[optional] 2. Software optimization
3. Energy efficient computing / power gating
4. Heterogenous computing
5. Multiprocessor
6. Advanced cooling

Suppose a new CPU,

85% capacitive load
15% voltage
15% frequency reduction
Find the increase in power ?

$$\frac{P_{new}}{P_{old}} = \frac{C \times \frac{85}{100} \left(V \times \frac{15}{100}\right)^2 F \times \frac{15}{100}}{C V^2 F} = 0.52$$

For processor A, clock rate = 4 GHz, voltage is 1.3V. static power is 20W and dynamic power is 110W.

(a) Find capacity

(b) Ratio of static and full power

(c) If dissipated power is reduced by 15% and current is same, what is the voltage?

(a) $f = 4 \times 10^9$ Hz, $V = 1.3$V, static power = 20 W, $P_d = 110$W

$$P_d = CV^2F$$

$$\Rightarrow 110 = C(1.3)^2 \times 4 \times 10^9$$

$$\Rightarrow C = 1.63 \times 10^{-8} \text{ F}$$

(b)

$$\frac{\text{Static}}{\text{Full}} = \frac{20}{(20+110)} \times 100 = 15.4\%$$

(c) New total power / Dissipated power $= 130 - \frac{15}{100} \times 130$

$= 110.5$

$$\text{Dissipated power} = P_{static} + P_{dynamic}$$

$$\Rightarrow 110.5 = x(VI) + C V_{new}^2 f$$

$$= x(VI) + C(xV)^2 f \quad [V_{new} = xV]$$

$$= 20x + x^2(CV^2f)$$

$$= 20x + x^2(110)$$

$$x = 0.9155$$

Multiprocessor

SPEC CPU Benchmark

↳ parallel processing

	Execution time	Reference time	SPEC ratio
Sudoko	698	1432	2.05
Chess	987	1705	1.73
Carom	1718	2939	1.71

↑

$$SPEC = \frac{ref}{ex}$$

$$\text{Overall geometric mean} = \sqrt[n]{2.05 \times 1.73 \times 1.71}$$

$$= \sqrt[3]{2.05 \times 1.73 \times 1.71}$$

Description	Name	Instruction Count x 10^9	CPI	Clock cycle time (seconds x 10^{-9})	Execution Time (seconds)	Reference Time (seconds)	SPECratio
Perl interpreter	perlbench	2684	0.42	0.556	627	1774	2.83
GNU C compiler	gcc	2322	0.67	0.556	863	3976	4.61
Route planning	mcf	1786	1.22	0.556	1215	4721	3.89
Discrete Event simulation - computer network	omnetpp	1107	0.82	0.556	507	1630	3.21
XML to HTML conversion via XSLT	xalancbmk	1314	0.75	0.556	549	1417	2.58
Video compression	x264	4488	0.32	0.556	813	1763	2.17
Artificial Intelligence: alpha-beta tree search (Chess)	deepsjeng	2216	0.57	0.556	698	1432	2.05
Artificial Intelligence: Monte Carlo tree search (Go)	leela	2236	0.79	0.556	987	1703	1.73
Artificial Intelligence: recursive solution generator (Sudoku)	exchange2	6683	0.46	0.556	1718	2939	1.71
General data compression	xz	8533	1.32	0.556	6290	6182	0.98
Geometric mean	–	–	–	–	–	–	2.36

Why geometric mean is used instead of average?

	comp A	comp B
P1	10	100
P2	100	10

↑

A as ref

B as ref

$$P1 = \frac{100}{10_A} = 10$$

$$P2 = \frac{10}{100_A} = 0.1$$

$$Avg = \frac{10 + 0.1}{2} = 5.05$$ (B is taking 5.05 time more than A) ✗

[or] $\sqrt{10 \times 0.1} = \sqrt{1} = 1$ ✓

The SPEC CPU2006 bzip2 benchmark on an AMD Barcelona processor has the following values:

- Instruction count = 2.389×10^{12}
- Execution time = 750 s
- Reference time = 9650 s

A new version of the processor has a 4 GHz clock rate. Its instruction count is reduced by 15%, execution time becomes 700 s, and the new SPECratio is 13.7. Find CPI

AMD,

$IC = 2.389 \times 10^{12}$

$T = 750s$

$Ref = 9650$

AMD_{new},

$IC = 2.89 \times 10^{12} - \left(\frac{15}{100} \times 2.89 \times 10^{12}\right) = 2.4 \times 10^{12}$

$f = 4 \times 10^9$ Hz

$T = 700s$

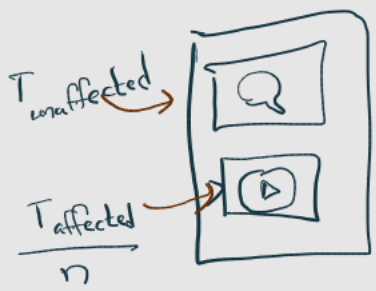
$SPECratio = 13.7$

$$CPU\ Time = \frac{IC \times CPI}{f}$$

$$\Rightarrow 700 = \frac{2.4 \times 10^{12} \times CPI}{4 \times 10^9}$$

$$\Rightarrow CPI = \square$$

Amdahl's Law



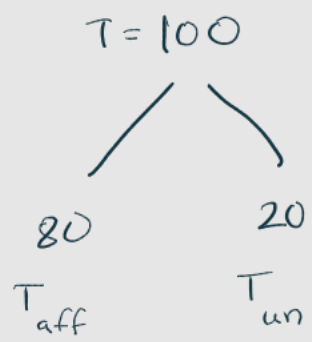
Make common case faster

$T_{improve}$

$$T_{improved} = \frac{T_{affected}}{\text{improvement factor}} + T_{unaffected}$$

How much improvement in multiply performance to get 2 time overall improvement? Multiply accounts for 80s/100s

$$T_{improved} = \frac{T_{affected}}{n} + T_{unaffected}$$



$$\Rightarrow \frac{100}{2} = \frac{80}{n} + 20$$

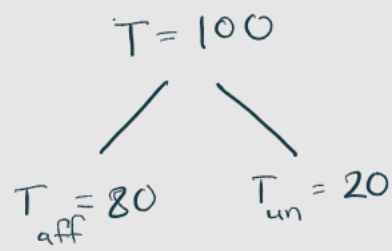
$$\Rightarrow n = \frac{8}{3}$$

How much improvement in multiply performance to get 5 time overall improvement? Multiply accounts for 80s/100s

$$\frac{100}{5} = \frac{80}{n} + 20$$

$$\Rightarrow 20 = \frac{80}{n} + 20$$

$$\Rightarrow n = \times$$



Task A = 35s ; Serial Task = 10s

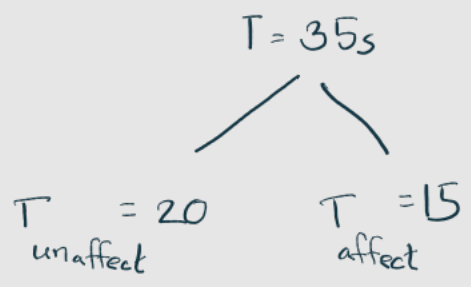
Parallel Task = 15s

To get 1.5 times performance how much should parallel task improve?

$$T_{improved} = \frac{T_{affected}}{n} + T_{unaffected}$$

$$\Rightarrow \frac{35}{1.5} = \frac{15}{n} + 20$$

↖ careful



CPU : 25%

GPU : 60%

I/O : 15%

Target speedup 3x

(a) Using Amdahl, prove if it is possible by changing GPU only?

(b) If I/O is 5 times faster, then is it possible by changing GPU?

(a)

$$T_{improved} = \frac{T_{affected}}{n} + T_{unaffected}$$

$$\Rightarrow \frac{100}{3} = \frac{60}{n} + 40$$

$$\Rightarrow n = \text{negative [Not possible]}$$

(b)

$$T_{improved} = \frac{T_{affected}}{n} + \frac{T_{affected}}{n} + T_{unaffected}$$

$$\Rightarrow \frac{100}{3} = \frac{60}{n} + \frac{15}{5} + (100 - 60 - 15)$$

Question - 12

You are optimizing the training pipeline for a new deep learning model. Profiling a single training epoch reveals the following execution time breakdown:

- Data Loading & Preprocessing (CPU): 25% of the total time
- Tensor Matrix Math (GPU): 60% of the total time
- Graph Evaluation & I/O: 15% of the total time

Your project advisor wants to reduce the total training time by a factor of 3 (an overall speedup of 3.0x). The advisor suggests taking the entire budget and buying a massive, state-of-the-art GPU cluster to accelerate *only* the Tensor Matrix Math.

a. Using Amdahl's Law, mathematically prove whether the advisor's goal of a 3.0x overall speedup is possible or impossible under this plan.

[Hint: $\frac{100}{3} = 25 + 15 + \frac{60}{n}$ *]*

b. If you invest in a multi-core CPU upgrade that successfully makes the Data Loading & Preprocessing phase 5 times faster, exactly how much faster must you make the GPU Tensor Matrix Math to hit your advisor's strict 3.0x overall speedup target?

[Hint: $\frac{100}{3} = \frac{25}{5} + 15 + \frac{60}{n}$ *]*

কেমনে বুঝবো Amphl থেকে Question এটা?

(1) Total time ভাগ হয়

(2) Improvement / speed up চায়

Consider a computer running a program that requires 250s, with 70s spent executing add instructions, 85s executed sub instructions, and 40s spent executing left-shift instructions.

a. By how much is the total time reduced if the time for add operations is reduced by 20%?
b. Can the total time can be reduced by 20% by reducing only the time for left-shift instructions?

(a) $T_{total} = \frac{T_{affected}}{n} + T_{unaffected}$

$= 56 + (250 - 70)$

$= 236s$

$70 - \frac{20}{100} \times 70 = 56$

(b) $236 - \left(\frac{20}{100} \times 236\right) = \frac{40}{n} + (250 - 40)$

$\Rightarrow$ n = negative

MIPS ← performance

Millions of Instruction Per Second

$$MIPS = \frac{\text{Instruction Count (IC)}}{\text{Execution time} \times 10^6}$$

$$= \frac{\cancel{\text{Instruction Count (IC)}}}{\frac{\cancel{IC} \times CPI}{\text{clock rate}} \times 10^6}$$

$$= \frac{\text{clock rate}}{CPI \times 10^6}$$

$$T = \frac{IC \times CPI}{f}$$

- add _____
- sub _____
- add _____
- sub _____
- sub _____
- mul _____
- sll _____

Computer A executes these instructions in 15s. Find MIPS

$$MIP = \frac{IC}{T \times 10^6} = \frac{7}{15 \times 10^6} \text{ Mips}$$

P1 has 3 GHz clock rate and CPI is 1.5. What is the performance?

✓ $f = 3 \times 10^9$ Hz

$CPI = 1.5$

$$IPS = \frac{\text{Clock rate}}{CPI \times 10^6} = \frac{3 \times 10^9}{1.5 \times 10^6} = 2 \times 10^9$$

A computer has MIPS of 5 and it takes 5s to execute a program. CPI is 2. (a) IC ? (b) Find the no of clock cycle.

(a) $MIPS = 5$

$$\Rightarrow \frac{IC}{\text{Execution time} \times 10^6} = 5$$

$$\Rightarrow \frac{IC}{5 \times 10^6} = 5$$

$$\Rightarrow IC = 25 \times 10^6$$

$MIPS = 5$

CPU Time = 5s

$CPI = 2$

(b) No of clock cycle = $IC \times CPI$